Project rules & grading

Semester Project – Neural Networks Laboratory

Objective

The goal of the semester project is to **solve a machine learning problem using neural network methods and techniques discussed in the course**.

The project should demonstrate:

- understanding of neural network models
- correct experimental methodology
- ability to analyze model behavior
- ability to interpret experimental results

Creativity and independent thinking are strongly encouraged.

Project Organization

Projects should be completed in:

- **groups of 2–4 students**

In exceptional cases, a **single-person project** may be allowed.

Project groups must include students from the same **laboratory group**.

The project will be developed **progressively during the semester**, with **periodic consultations during laboratory meetings**.

Final Project Deliverables

The final project must include the following components.

1. Implementation

The project must include **working code**, preferably in the form of:

- Jupyter Notebook (.ipynb)

The code must:

- run from top to bottom without errors
- use fixed random seeds where appropriate
- produce reproducible results

A short **README** explaining how to run the project is recommended.

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2. Experiments and Results

The project must include:

- results of conducted experiments
- intermediate and final results
- comparison between different model configurations (if applicable)

Experiments must be **reproducible**.

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3. Analysis

The project should clearly demonstrate:

- understanding of the problem
- interpretation of model behavior
- discussion of obtained results
- identification of limitations and possible improvements

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4. Final Presentation

Each group will give a **short final presentation** during the last laboratory meetings.

The presentation should include:

- problem description
- methodology
- key experiments
- results and conclusions

Students must also be prepared to **answer questions about their implementation and experiments**.

Project Evaluation by Milestones

The project will be evaluated progressively during the semester through four project meetings.

Stage	Points
Project topic meeting	6
First progress meeting	12
Second progress meeting	14
Final presentation	8

Total: 40 points

Important Requirement

The semester project is **mandatory**.

To pass the laboratory course:

> A student must obtain **at least 50% of the project points (20 / 40)**.

Obtaining fewer than **20 project points results in automatic failure of the laboratory**, regardless of assignment points.

Project Topics

- object detection
- semantic segmentation
- text classification

Project Timeline

The project will progress during the semester through **four project meetings**.

1. Project Topic Meeting (6 points) - March 31

Evaluation is based on:

- clarity of the problem definition
- suitability of the chosen dataset
- feasibility of the project scope

- quality of the proposed approach

Students should present the **problem they want to solve and the planned methodology**.

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2. First Progress Meeting (12 points) - April 28

Evaluation is based on:

- correct formulation of the ML task
- implementation of a baseline solution
- initial experiments
- correctness of the experimental setup

Students should demonstrate **that the project pipeline is already working**.

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3. Second Progress Meeting (14 points) - May 26

Evaluation is based on:

- quality and scope of conducted experiments
- improvements over the baseline model
- comparison between approaches
- interpretation and analysis of results

At this stage the project should be **close to completion**.

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4. Final Presentation (8 points) - June 16

Evaluation is based on:

- clarity and structure of the presentation
- quality of the final conclusions
- interpretation of experimental results
- ability to answer questions about the project